MTH 310 **Final Exam** Name:____________________

Spring, 2026 Calculus III (Fall 2025)

**Instructions.** (200 points) Show work for credit. Calculator OK (but not useful).

(18pts) **1.** Consider the line that passes through the point $(2, 0, 8)$ and is orthogonal to the plane $2x + 4y - 3z = 9$. Find the point at which this line intersects the plane.

**Solution:**

The line is orthogonal to the plane, so it has direction vector equal to the plane's normal:

$$\mathbf{n} = \langle 2, 4, -3 \rangle.$$

A parametrization is

$$\ell(t) = (2, 0, 8) + t\langle 2, 4, -3 \rangle = (2 + 2t,\ 4t,\ 8 - 3t).$$

Impose the plane equation $2x + 4y - 3z = 9$:

$$2(2 + 2t) + 4(4t) - 3(8 - 3t) = 9 \ \Rightarrow\ 29t - 20 = 9 \ \Rightarrow\ t = 1.$$

The intersection point is $\boxed{(4, 4, 5)}$.

18 pts

(18[pts]) **2.** Show that the limit $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ is zero along every line of the form $y = mx$ with $m \neq 0$. Just because the limit is zero along all lines, does that prove the limit is zero? Prove it is zero or demonstrate that it isn't.

**Solution:**

Along any line $y = mx$ with $m \neq 0$:

$$\frac{x^2(mx)}{x^4+(mx)^2} = \frac{mx^3}{x^2(x^2+m^2)} = \frac{mx}{x^2+m^2} \to 0 \quad \text{as } x \to 0.$$

So the limit is 0 along every such line (and also along $y = 0$).
However, along the parabola $y = x^2$:

$$\frac{x^2y}{x^4+y^2} = \frac{x^2 \cdot x^2}{x^4+x^4} = \frac{x^4}{2x^4} = \tfrac{1}{2},$$

so the function tends to $\frac{1}{2}$ along that path, not 0. Therefore the two-variable limit at $(0,0)$ **does not exist**.

18 pts

(18$^{\text{pts}}$) **3.** Suppose a mountain can be represented by the function $f(x, y) = x^2 + 2y^2 + xy - 2y$.

(a) *(9 pts)* If you are on the mountain at the point $(0, 1, 0)$, in what direction should you move in order to decrease in elevation most quickly?

**Solution:**

$\nabla f(x, y) = \langle 2x + y,\ 4y + x - 2\rangle$. At $(0, 1)$: $\nabla f(0, 1) = \langle 1, 2\rangle$.

The direction of most rapid decrease is the negative gradient, normalized:

$$\mathbf{u}_{\text{decrease}} = -\frac{\nabla f(0,1)}{\|\nabla f(0,1)\|} = -\frac{\langle 1, 2\rangle}{\sqrt{5}} = \boxed{\left\langle -\tfrac{1}{\sqrt{5}},\ -\tfrac{2}{\sqrt{5}} \right\rangle}.$$

(b) *(9 pts)* If you are at the point $(0, 1, 0)$ and decide to walk toward the point $(4, 4, 56)$, then what is the rate of change of elevation as you start your walk?

**Solution:**

The direction of motion in the $xy$-plane is from $(0, 1)$ to $(4, 4)$:

$$\mathbf{v} = \langle 4, 3\rangle, \quad \|\mathbf{v}\| = 5, \quad \mathbf{u} = \left\langle \tfrac{4}{5},\ \tfrac{3}{5} \right\rangle.$$

The rate of change of elevation is the directional derivative

$$D_{\mathbf{u}} f(0, 1) = \nabla f(0, 1) \cdot \mathbf{u} = \langle 1, 2\rangle \cdot \left\langle \tfrac{4}{5}, \tfrac{3}{5} \right\rangle = \frac{4 + 6}{5} = \boxed{2}.$$

18 pts

(18pts) **4.** Give an equation for the tangent plane to the surface $x^2z^3 + 3y^3z = x^3\cos(\pi y) + 9$ at the point $(1, -1, 1)$.

**Solution:**

Let $F(x, y, z) = x^2z^3 + 3y^3z - x^3\cos(\pi y) - 9 = 0$. The tangent plane uses the normal $\nabla F(1, -1, 1)$. Compute the partials:

$$F_x = 2xz^3 - 3x^2\cos(\pi y), \quad F_y = 9y^2z + x^3\pi\sin(\pi y), \quad F_z = 3x^2z^2 + 3y^3.$$

At $(1, -1, 1)$:

$$F_x = 2(1)(1) - 3(1)\cos(-\pi) = 2 - 3(-1) = 5,$$
$$F_y = 9(1)(1) + 1 \cdot \pi\sin(-\pi) = 9,$$
$$F_z = 3(1)(1) + 3(-1) = 0.$$

A normal vector is $\langle 5, 9, 0\rangle$, so the tangent plane is

$$5(x-1) + 9(y+1) + 0(z-1) = 0 \implies \boxed{5x + 9y + 4 = 0}.$$

18 pts

(20pts) **5.** Find all absolute maxima and minima of $f(x, y) = 2x^2 + 3y^2 - 4x - 5$ over the region $x^2 + y^2 \leq 4$. Don't forget to locate and check the critical points along the boundary.

**Solution:**

**Interior critical points.** $\nabla f = \langle 4x - 4,\ 6y \rangle = \mathbf{0}$ gives $x = 1, y = 0$, and $(1, 0)$ is inside $x^2 + y^2 \leq 4$:

$$f(1, 0) = 2 - 4 - 5 = -7.$$

**Boundary $x^2 + y^2 = 4$.** Parametrize $x = 2\cos t$, $y = 2\sin t$:

$$f = 8\cos^2 t + 12\sin^2 t - 8\cos t - 5 = -4\cos^2 t - 8\cos t + 7.$$

Let $u = \cos t \in [-1, 1]$: $f(u) = -4u^2 - 8u + 7$, $f'(u) = -8u - 8 = 0$ at $u = -1$. Concave down, so on $[-1, 1]$:

$$f(-1) = -4 + 8 + 7 = 11, \qquad f(1) = -4 - 8 + 7 = -5.$$

**Compare:**

$$\boxed{\text{Absolute min} = -7 \text{ at } (1, 0); \quad \text{Absolute max} = 11 \text{ at } (-2, 0).}$$

20 pts

(18[pts]) **6.** Evaluate $\iiint_E z\,dV$ where $E$ is the solid bounded above by $x^2+y^2+z^2=1$ and below by $z=\sqrt{x^2+y^2}$.

**Solution:**

The cone $z=\sqrt{x^2+y^2}$ corresponds to $\phi=\pi/4$ in spherical coordinates; the sphere is $\rho=1$. So

$$0\le\rho\le1,\quad 0\le\phi\le\tfrac{\pi}{4},\quad 0\le\theta\le2\pi.$$

With $z=\rho\cos\phi$ and $dV=\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$:

$$\iiint_E z\,dV=\int_0^{2\pi}\int_0^{\pi/4}\int_0^1\rho^3\cos\phi\sin\phi\,d\rho\,d\phi\,d\theta.$$

The integral separates:

$$=2\pi\cdot\int_0^{\pi/4}\cos\phi\sin\phi\,d\phi\cdot\int_0^1\rho^3\,d\rho=2\pi\cdot\tfrac{1}{4}\cdot\tfrac{1}{4}=\boxed{\tfrac{\pi}{8}}.$$

18 pts

(18pts) **7.** Evaluate $\int_C x\,ds$ where $C$ is the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$.

**Solution:**

Parametrize: $x = t$, $y = t^2$, $0 \le t \le 1$. Then

$$ds = \sqrt{1+4t^2}\,dt, \qquad \int_C x\,ds = \int_0^1 t\sqrt{1+4t^2}\,dt.$$

Let $u = 1+4t^2$, $du = 8t\,dt$, so $t\,dt = \frac{1}{8}\,du$, with $u$ from 1 to 5:

$$\int_0^1 t\sqrt{1+4t^2}\,dt = \frac{1}{8}\int_1^5 u^{1/2}\,du = \frac{1}{8}\cdot\frac{2}{3}\big[u^{3/2}\big]_1^5 = \frac{1}{12}\big(5\sqrt{5}-1\big).$$

$$\boxed{\int_C x\,ds = \frac{5\sqrt{5}-1}{12}.}$$

18 pts

(18pts) **8.** Compute $\int_C (2x + \sin y)\,dx + (x\cos y - 6y)\,dy$, where $C$ consists of line segments from $(0,0)$ to $(2,1)$ and from $(2,1)$ to $(0,2)$.

**Solution:**

Let $P = 2x + \sin y$, $Q = x\cos y - 6y$. Check: $P_y = \cos y = Q_x$, so $\mathbf{F}$ is conservative on $\mathbb{R}^2$.

**Find a potential.** From $f_x = 2x + \sin y$: $f = x^2 + x\sin y + g(y)$. Match $f_y = x\cos y + g'(y) = x\cos y - 6y$, so $g'(y) = -6y$ and $g(y) = -3y^2$:

$$f(x,y) = x^2 + x\sin y - 3y^2.$$

**Apply FTLI.** The path goes from $(0,0)$ to $(0,2)$ overall (the middle point doesn't matter for a conservative field):

$$\int_C P\,dx + Q\,dy = f(0,2) - f(0,0) = (0 + 0 - 12) - 0 = \boxed{-12}.$$

18 pts

(18pts) **9.** Let $C$ be the circle of radius 5 centered at the origin traversed counterclockwise. Compute

$$\int_C (e^{\sin x} - y^3)\,dx + x^3\,dy.$$

*Hint: computing this directly isn't the way to go...*

**Solution:**

Use Green's Theorem on the disk $D$ of radius 5. With $P = e^{\sin x} - y^3$ and $Q = x^3$:

$$Q_x = 3x^2, \quad P_y = -3y^2, \quad Q_x - P_y = 3(x^2 + y^2).$$

By Green's,

$$\oint_C P\,dx + Q\,dy = \iint_D 3(x^2 + y^2)\,dA.$$

In polar with $0 \le r \le 5$:

$$\iint_D 3(x^2 + y^2)\,dA = 3\int_0^{2\pi}\int_0^5 r^2 \cdot r\,dr\,d\theta = 3 \cdot 2\pi \cdot \tfrac{5^4}{4} = \boxed{\tfrac{1875\pi}{2}}.$$

18 pts

(18pts) **10.** Let $S$ be the surface parameterized by $\mathbf{r}(u,v) = \langle u\cos v,\ u\sin v,\ v\rangle$ for $0 \le u \le 2$, $0 \le v \le \pi$. Compute $\iint_S \sqrt{x^2+y^2}\,dS$.

**Solution:**

On the surface, $x^2+y^2 = u^2\cos^2 v + u^2\sin^2 v = u^2$, so the integrand is $\sqrt{x^2+y^2} = u$. Compute the surface element:

$$\mathbf{r}_u = \langle \cos v,\, \sin v,\, 0\rangle, \quad \mathbf{r}_v = \langle -u\sin v,\, u\cos v,\, 1\rangle,$$

$$\mathbf{r}_u \times \mathbf{r}_v = \langle \sin v,\, -\cos v,\, u\rangle, \quad \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{1+u^2}.$$

Therefore

$$\iint_S \sqrt{x^2+y^2}\,dS = \int_0^\pi \int_0^2 u\sqrt{1+u^2}\,du\,dv = \pi \int_0^2 u\sqrt{1+u^2}\,du.$$

With $w = 1+u^2$, $dw = 2u\,du$:

$$\pi \cdot \tfrac{1}{2}\int_1^5 w^{1/2}\,dw = \pi \cdot \tfrac{1}{3}\left(5^{3/2}-1\right) = \boxed{\tfrac{\pi}{3}\left(5\sqrt{5}-1\right)}.$$

18 pts

(18pts) **11.** Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle x^3, y^3, y^3 \rangle$ and $S$ is the surface of the cylindrical solid bounded by $x^2 + y^2 = 1$, $z = 0$, $z = 4$, with outward orientation.

**Solution:**

By the Divergence Theorem, $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{F}\, dV$ where $V$ is the solid cylinder.

Compute the divergence:

$$\nabla \cdot \mathbf{F} = \tfrac{\partial}{\partial x}(x^3) + \tfrac{\partial}{\partial y}(y^3) + \tfrac{\partial}{\partial z}(y^3) = 3x^2 + 3y^2 + 0 = 3(x^2 + y^2).$$

In cylindrical coordinates with $0 \le r \le 1$, $0 \le \theta \le 2\pi$, $0 \le z \le 4$:

$$\iiint_V 3(x^2 + y^2)\, dV = \int_0^{2\pi} \int_0^1 \int_0^4 3r^2 \cdot r\, dz\, dr\, d\theta = 3 \cdot 2\pi \cdot 4 \cdot \tfrac{1}{4} = \boxed{6\pi}.$$

18 pts